

MAHER MOSTAFA

Backend Software Engineer

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Summary

Backend Software Engineer specializing in Laravel, PHP, and PostgreSQL. Experienced in building and maintaining production backend systems for financial workflows, payment integrations, multi-tenant applications, and operational platforms, with a strong focus on data correctness, reliable processing, and maintainable service boundaries.

Skills

TECHNICAL SKILLS:

- **Languages & Frameworks:** PHP 8.2–8.4, Laravel 11–13, Hypervel/Swoole, Laravel Octane/RoadRunner
- **Data & APIs:** PostgreSQL, MySQL, Redis, SQL, REST API design, PostgreSQL JSONB, CTEs, partial indexes, schema-per-tenant design
- **Architecture:** Multi-tenancy, service/repository layers, DTOs, domain enums, compensating transactions, ledger design, Composer package extraction
- **Payments & Security:** Wallet ledgers, payment gateway abstraction, webhook idempotency, transaction rollback, API-key authentication, bcrypt key rotation, HMAC verification, rate limiting

Experience

Storage Station Cairo, Egypt

On-site (Present)

Backend engineering — systems owned end-to-end

Hypervel Billing Service | Financial microservice · Hypervel/Swoole, PostgreSQL

Jan 2026

- Implemented schema-per-tenant PostgreSQL storage and dynamic connection registration, separating the platform registry from tenant billing schemas.
- Built transaction-safe wallet operations with lock checks and conditional balance updates for recharge, conversion, and debit flows.
- Added linked ledger reversal records for auditable financial corrections, and deduplicated PayTabs callbacks by charge ID before wallet credit; verified callbacks with HMAC.
- Unified four payment providers behind the private hypervel/payments package, separating charge creation, webhook handling, and refunds from billing-domain code.
- Built PostgreSQL CTE-based invoicing and credit reports with unit-tested query logic; delivered Docker, health checks, GitHub Actions test gating, semver tagging, and AWS ECR image publishing.

Survey Service — integration platform (Laravel 13, PostgreSQL, Filament)

Jun 2026

- Built a configuration-driven application → module → event resolution flow so integrations can be added without client-specific code paths.
- Enforced idempotent survey assignment in both the service layer and PostgreSQL partial unique indexes.
- Separated authentication by trust boundary: X-App-Key for server-to-server APIs and HMAC-SHA256 signed tokens for public survey links.
- Implemented metadata-based survey routing with six condition operators, model-level validation for mutually exclusive scopes, and controllers → services → repositories structure; added 41 automated tests for API flows, token verification, and scope validation.

Packing Item System | Reusable Laravel package

Feb 2026

- Extracted packing-materials logic into a versioned Composer package with a trait-composed service and Facade API for reuse across production applications.
- Preserved historical order values by snapshotting unit and total prices at checkout; split carts into per-carrier orders within database transactions to match fulfillment and billing boundaries.
- Added polymorphic request/response audit records for carrier integrations without coupling domain models to carrier APIs.

Bolesa Egypt — Backend Team Contributor | Production Platform: bolesa.net

- Built payment and billing backend flows that generate pending-order payment links, convert successful PaymentCaptured events into AWBs with issuing-fee ledger entries, and accrue reward points for MADA payments and approved bank transfers.
- Developed shipping AWB lifecycle automation with Horizon-queued carrier-status synchronization and automatic cancellation/refund-request handling when AWBs are deleted or cancelled.
- Extended the config-driven storageitsolutions/api-tables admin layer to support plan/carrier pricing management, payment-transaction observability, and shipping-AWB operational actions.
- Delivered scheduled Excel operational reports for customer invoicing, monthly credits, and points conversion through SendCustomReportEmailJob on Horizon’s reports queue.
- Improved production integrations by adding SPL National Address and HERE short-address lookup APIs with credit validation, and by hardening Salla, Salla-plugin, and Shopify order-ingestion against stuck unique-constraint failures.
- Implemented bank-account change and customer document workflows with supporting-document uploads, temp_bank_updates API-table handling, migrations, and customer/admin notifications.

CareQ — salon booking platform

May 2025

- Built concurrent-safe slot reservation with expiry and role-separated authentication using Sanctum and OTP; used PostgreSQL/PostGIS for serviceable-area constraints and Laravel Octane/RoadRunner.

3Hand Design & Marketing Solutions — Backend Engineer

Mar 2023

- Modeled financial-domain APIs for safes, banks, branches, and account statuses with validation-heavy business rules in the service layer.
- Built server-side validation for dynamic purchase and expense workflows used by AJAX admin forms.

EPIK Software — Backend Engineer

Remotely | Oct 2023

- Maintained authenticated REST APIs with RBAC authorization and MySQL schema/query optimization;participated in API contract design and code review for frontend integration.

Additional Production Work

- **Goods Platform (3Hand):** Filament admin, RBAC, and Paymob subscriptions. [goods.eg](#)
- **Inventory Management (EPIK):** purchase/sales order and stock tracking APIs

Sept 2024
Apr 2024

Education

Bachelor’s Degree, Computer Engineering — EL-Shorouk Academy

Grade: Good | 2019–2023

- Relevant coursework: Data Structures, Object-Oriented Programming, Algorithm Analysis, Software Engineering, Operating Systems, Computer Architecture, Compiler Theory, Artificial Intelligence.

Training / Courses

- Embedded Systems — Information Technology Institute (ITI) · Back-End Development — Itrax Academy